Azure Networking Practical Guide

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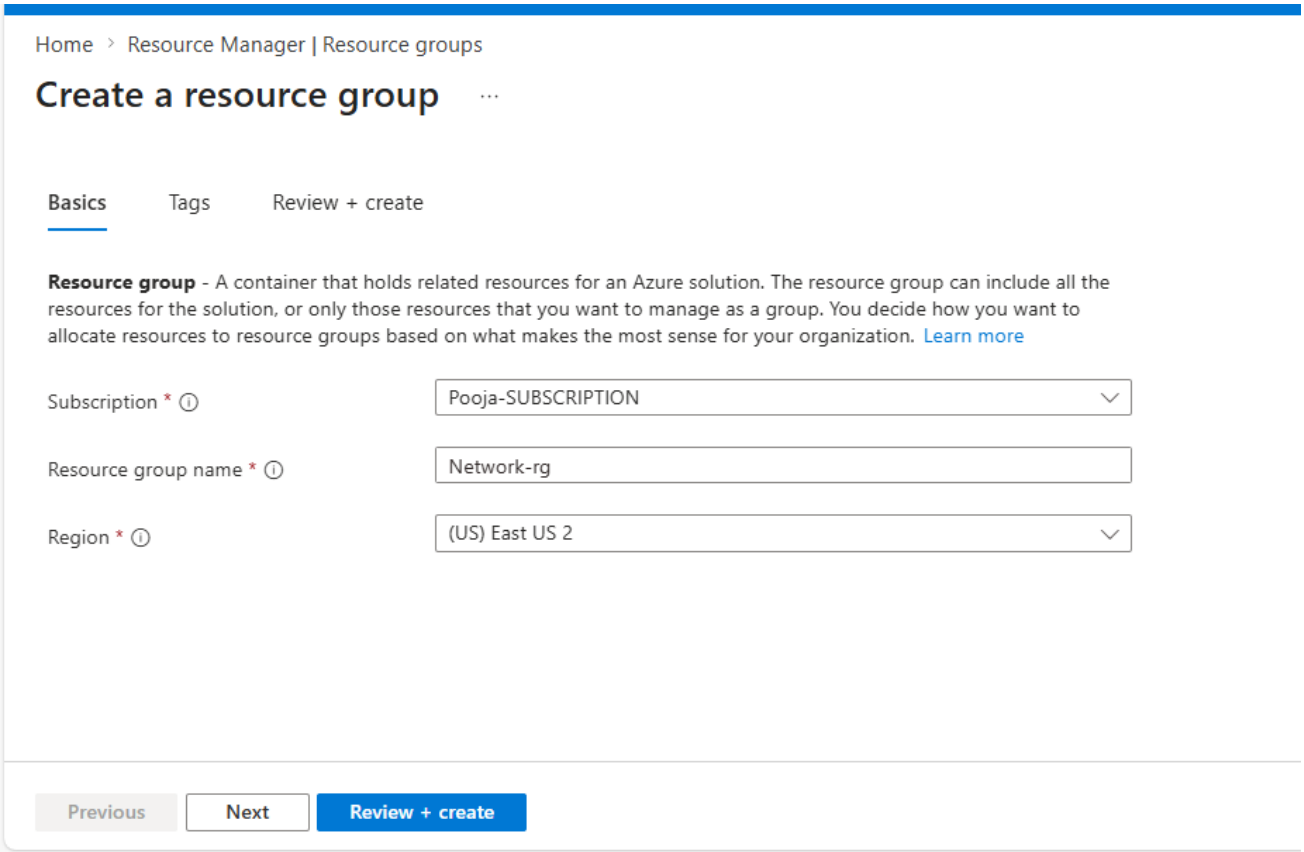
Create Your resource group

- Go to the Azure portal https://portal.azure.com and sign in.
- In the top search bar, type Resource groups and open it.
- Click on Create.
- Select your subscription.
- Enter a Resource group name (Ex: Network-rg).
- Choose a region (for example: East US 2).



Click **Review + Create**, wait for validation, then click **Create**.

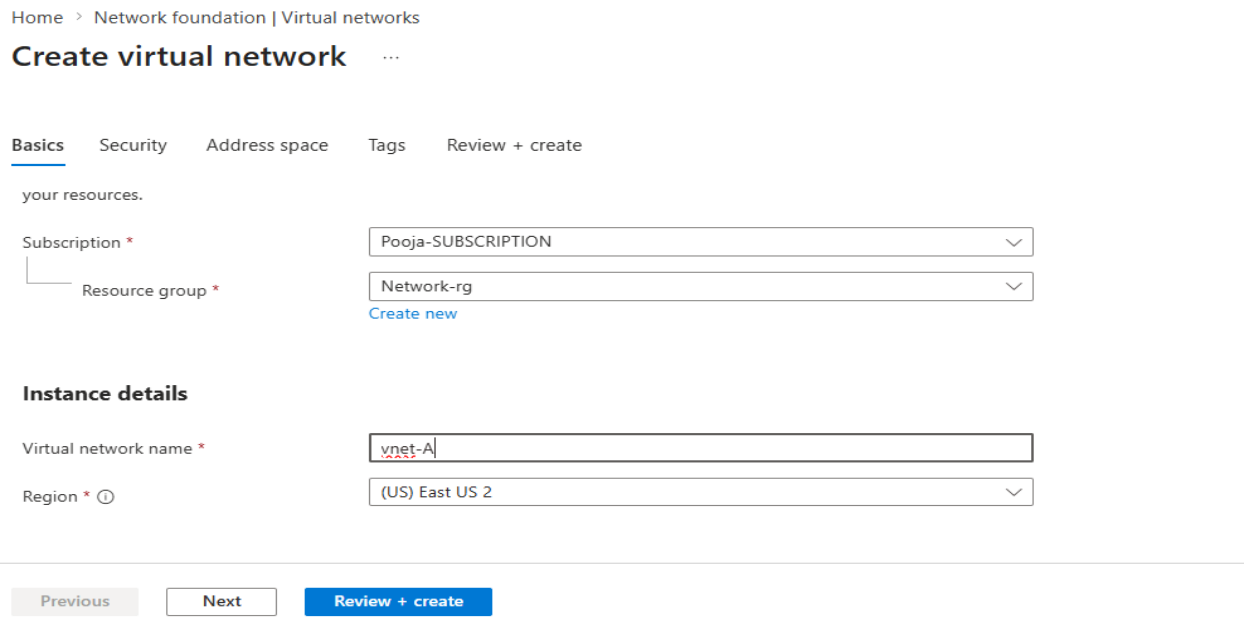
Note: Use the same subscription, region, and resource group that you create for this experiment, and use them consistently for all resources.

Exercise 1 – vnet creation + vnet - vnet peering

Step 1: Create first VNet

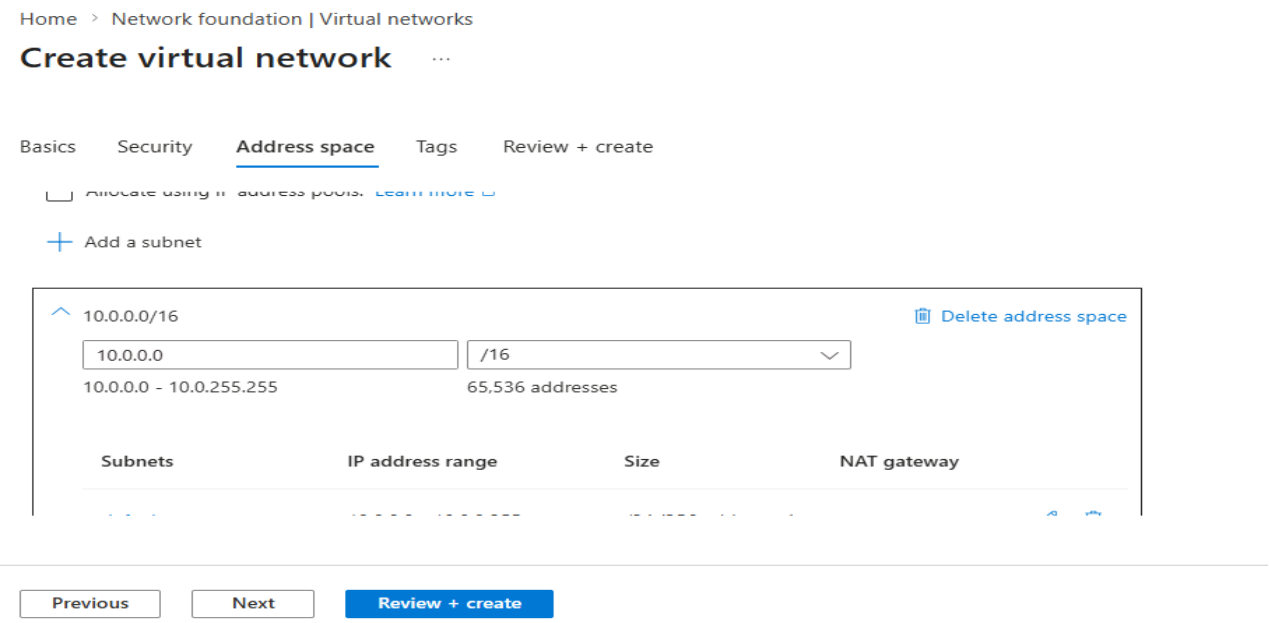
Azure Portal → Search for Virtual Networks
Click + Create
Fill:

- **Subscription:** Choose your subscription
- **Resource Group**: Choose your RG
- **Name**: Give any name (ex: vnet-A)
- **Region**: Same as resource group



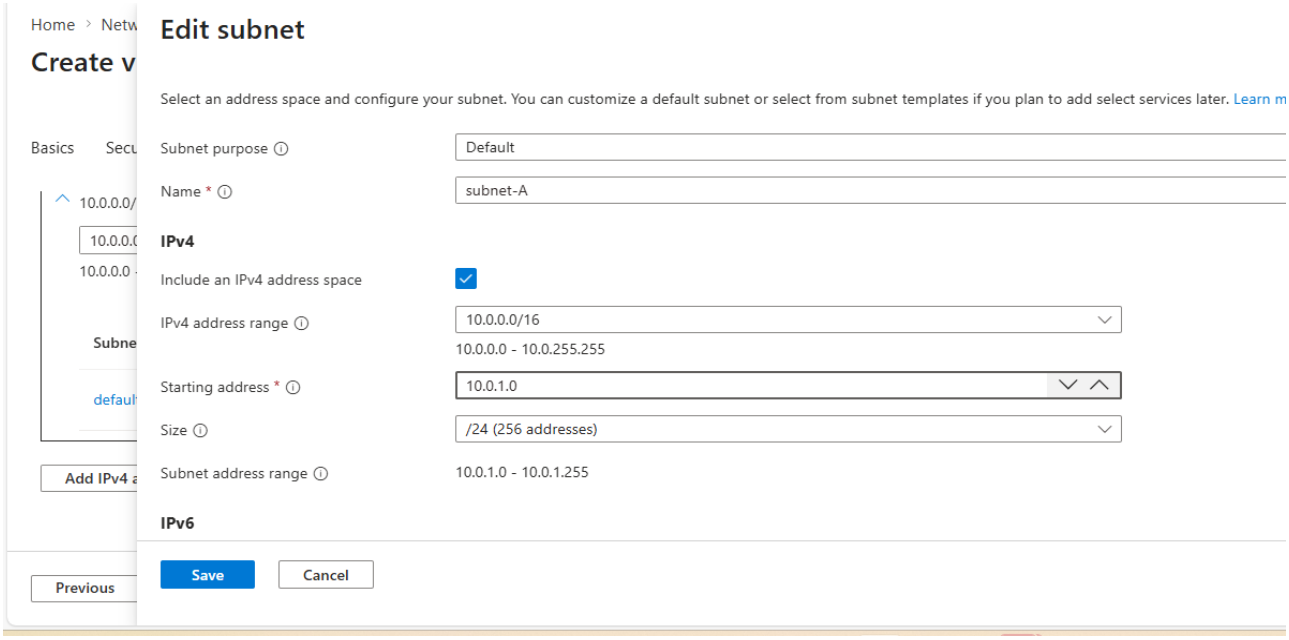
Go to address space tab

- **Address space**: 10.0.0.0/16



Next click on **default subnet** and edit it

- **Subnet name**: Subnet-A
- **Subnet range**: 10.0.1.0/24



Click save

Click **Review + Create**, wait for validation, then click **Create**.

Step 2: Create second VNet

Same steps as above with:

- **Name**: vnet-B
- **Address space**: 10.1.0.0/16
- **Subnet:** Subnet-B
- **Subnet range:** 10.1.1.0/24

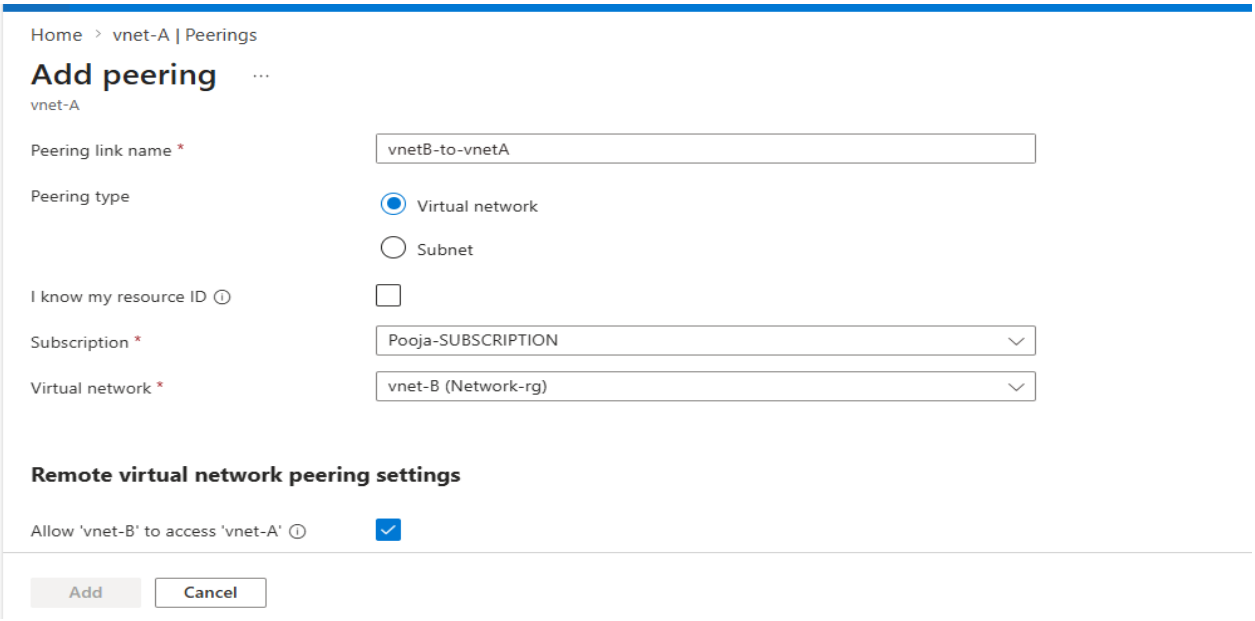
Address spaces must NOT overlap.

Step 3: Create VNet Peering (A → B)

1. Open **vnet-A**
2. Click Left Navigation Pane > **Setting** > **Peerings**
3. Click **+ Add**
4. Fill:

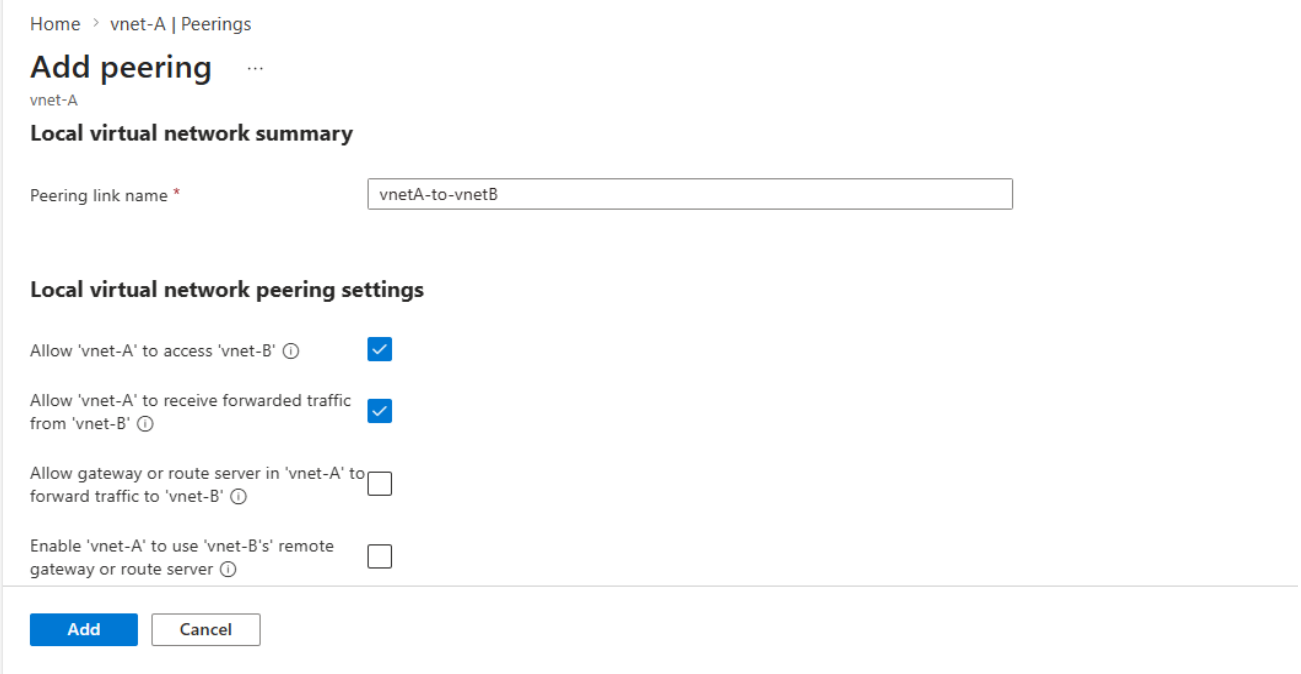
Remote virtual network (vnet-B side)

- Peering link name: vnetB-to-vnetA
- Peering type: Virtual Network
- Subscrption: Choose yours
- Virtual Network: Remote network vnet-B
- Allow vnet-B to access vnet-A → **Enable**
- Allow vnet-B to receive forwarded traffic from vnet-A → **Enable**



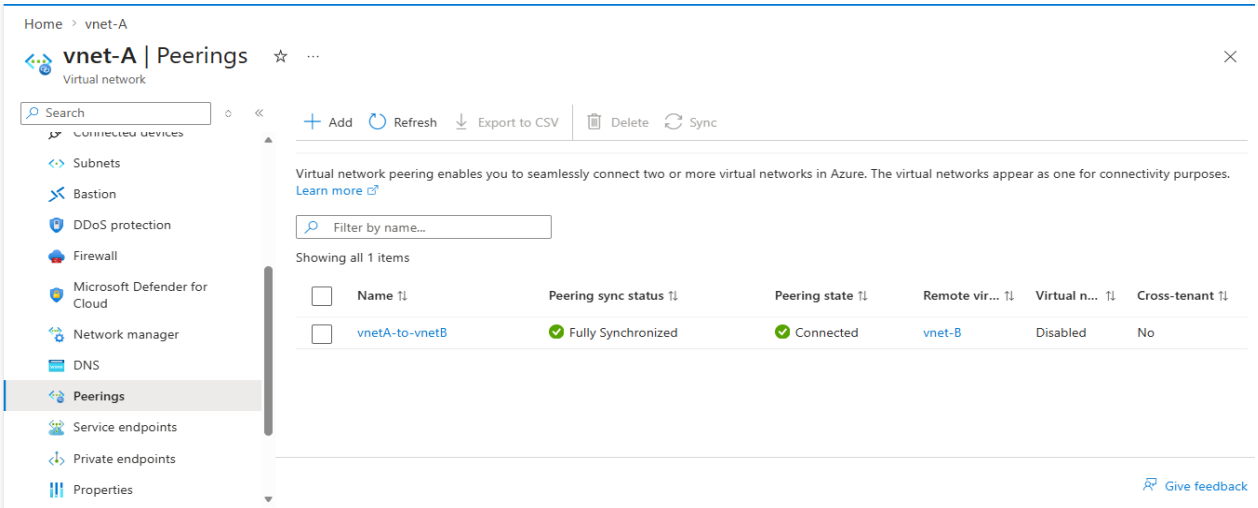
Local virtual network (vnet-A side)

- Peering link name: vnetA-to-vnetB
- Allow vnet-A to access vnet-B → Enable
- Allow vnet-B to receive forwarded traffic from vnet-A → **Enable**



5. Click **Add**

Now check peering added or not



Go to vnet-B and verify the peering -- it will be created automatically when both VNets are in the same subscription and you configured both sides during peering creation. No manual configuration is required on vnet-B.

Step 4. Verify vnet – vnet peering

Create VM in VNet-A

1. Go to **Virtual Machines**
2. Click + **Create**
3. Basics:
    - Name: vnetA-vm
    - Availability option: No infrastructure redundancy required
    - Security type: keep default
    - VM architecture: keep default
    - Image: Ubuntu server
    - Size: basic (if not available choose standard)
    - Authentication type: Password
    - Public inbound ports: Allow selected ports
    - Select inbound ports: ssh
4. Networking:
    - Virtual Network: vnet-A
    - Subnet: subnet-A
5. Remaining keep as it is, Click **Create**

Create VM in VNet-B

1. Follow same steps as above
2. Basics:
    - Name: vnetB-vm
3. Networking:
    - Virtual Network: vnet-B
    - Subnet: subnet-B
4. Remaining keep as it is, Click **Create**

Now connect to VM using SSH

1. Go to vnetA-vm
2. In left pane **Connect > Connect**
3. Copy **ssh** command and paste it in command prompt

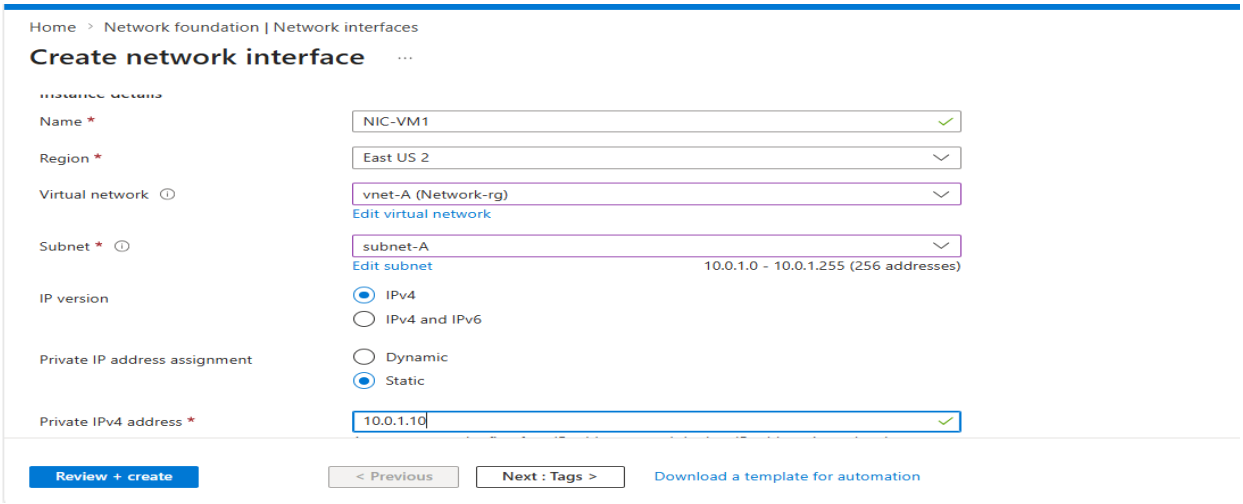
4. Are you sure you want to continue connecting? **yes**
5. Give your vm password when it asks
6. Go to vnetB-vm scroll down **Properties** > **Networking** > **Copy VM private ip**
7. Now ping VnetB-vm ip from VnetA-vm using following comand
    - Ping < vnetA-vm private ip >
8. If you get following ouput your peering is working

```
AZUREUSER@vnetA-vm: ~
AZUREUSER@vnetA-vm:~$ ping 10.0.1.7
PING 10.0.1.7 (10.0.1.7) 56(84) bytes of data.
64 bytes from 10.0.1.7: icmp_seq=1 ttl=64 time=1.55 ms
64 bytes from 10.0.1.7: icmp_seq=2 ttl=64 time=0.731 ms
64 bytes from 10.0.1.7: icmp_seq=3 ttl=64 time=1.66 ms
64 bytes from 10.0.1.7: icmp_seq=4 ttl=64 time=0.893 ms
64 bytes from 10.0.1.7: icmp_seq=5 ttl=64 time=0.807 ms
64 bytes from 10.0.1.7: icmp_seq=6 ttl=64 time=0.915 ms
64 bytes from 10.0.1.7: icmp_seq=7 ttl=64 time=0.955 ms
64 bytes from 10.0.1.7: icmp_seq=8 ttl=64 time=3.40 ms
64 bytes from 10.0.1.7: icmp_seq=9 ttl=64 time=0.771 ms
^C
--- 10.0.1.7 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8115ms
rtt min/avg/max/mdev = 0.731/1.298/3.403/0.809 ms
AZUREUSER@vnetA-vm:~$
```

Exercise 2 – assign static private ip to VM

Step 1: Create NIC with Static IP

1. Search **Network Interfaces**
2. Click **+ Create**
3. Enter:
    - Name: NIC-VM1
    - VNet: VNet-A
    - Subnet: Subnet-A
    - IP version: ipv4
    - Private IP address assignment: static
    - IP address: 10.0.1.10



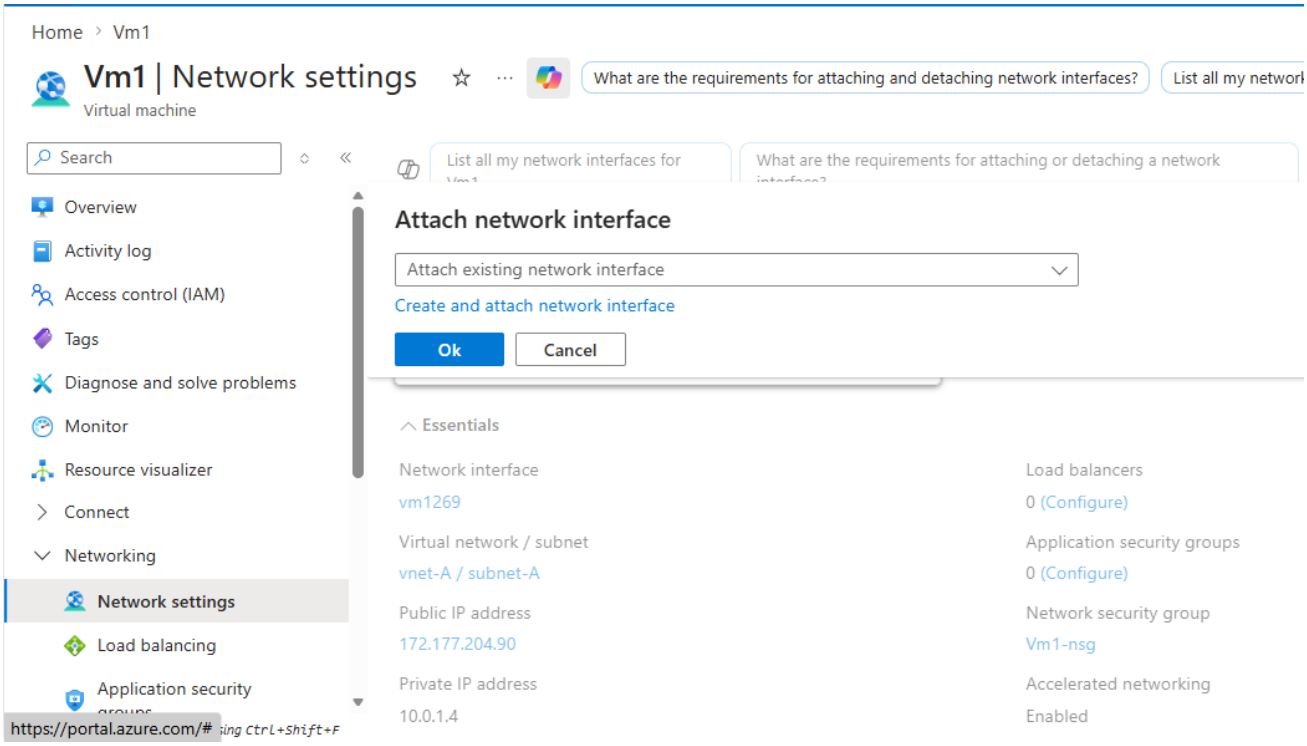
4. Click **Review + Create**, wait for validation, then click **Create**. VM1 now has a **static private IP**

Step 2: Create VM and Attach NIC

1. Go to **Virtual Machines**
2. Click **+ Create**
3. Basics:
    - Name: VM1
    - Availability option: No infrastructure redundancy required
    - Security type: keep default
    - VM architecture: keep default
    - Image: Ubuntu server
    - Size: Standard
    - Authentication type: Password
    - Public inbound ports: Allow selected ports
    - Select inbound ports: ssh
4. Click **Create**

Step 3: Attach NIC to VM

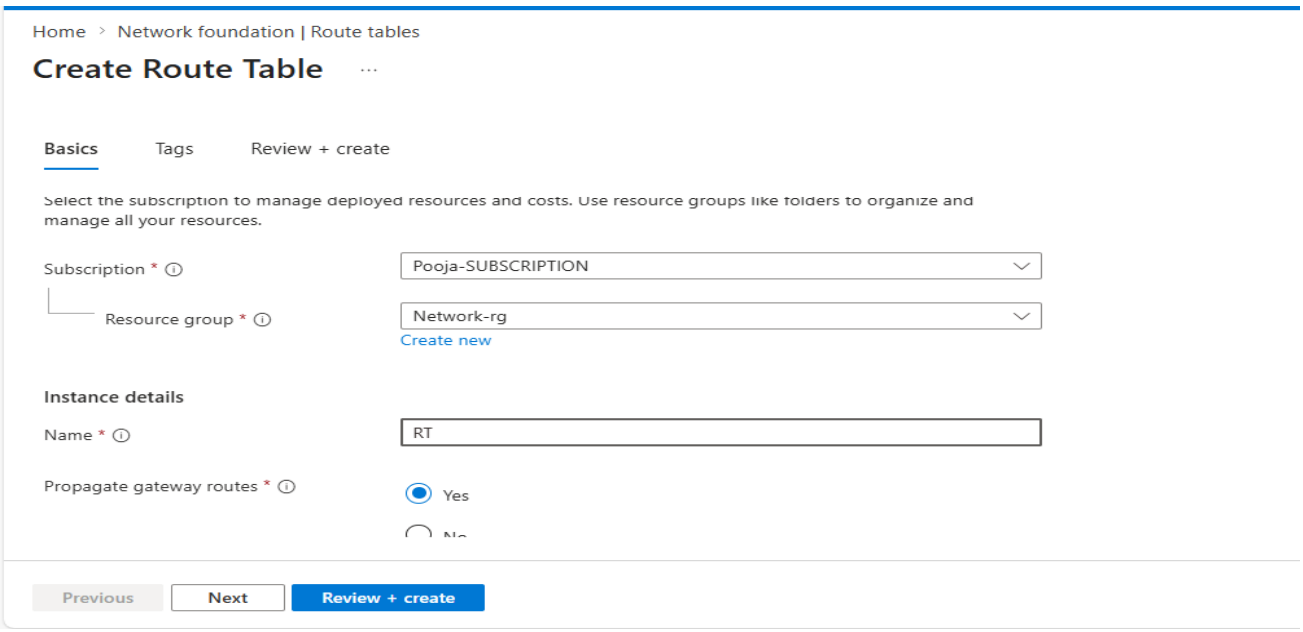
- Go to **Virtual Machines** → select your VM
- Click **Stop (Deallocate)** and wait until it fully stops
- In VM menu, open **Networking > Network setting**
- Click **Attach network interface**
- Select your **existing NIC**
- Click **OK**
- Go to **overview** and Start the VM again



Exercise 3 – route table & custom routes

Step 1: Create Route Table

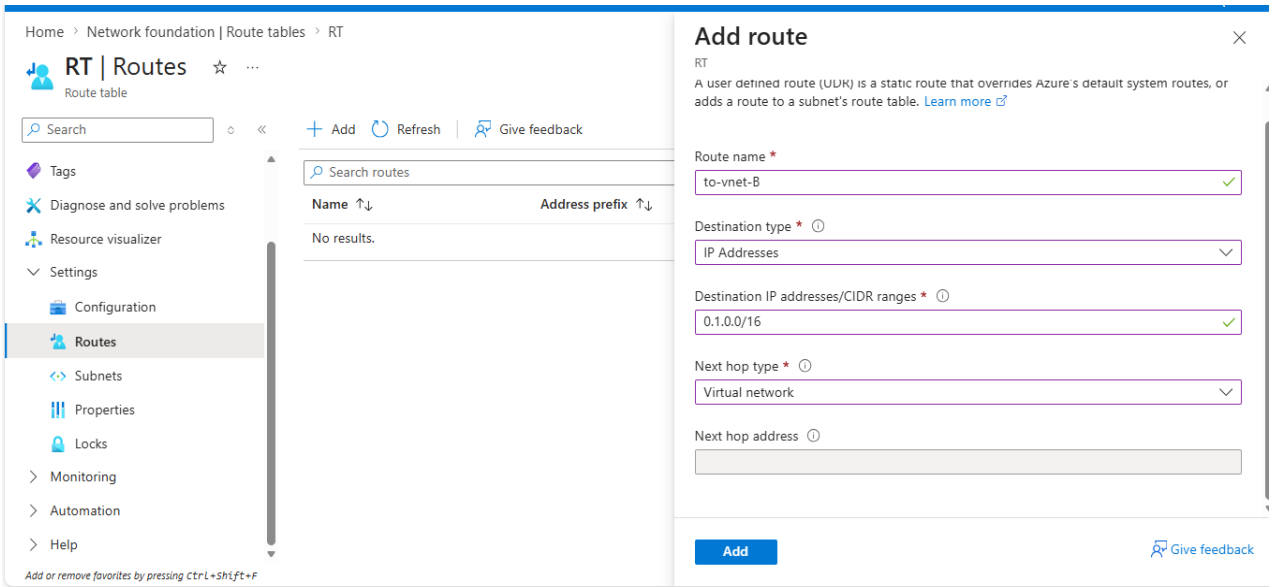
1. In portal search **Route Tables**
2. Click **+ Create**
3. Name: RT
4. Resource Group: Choose your RG



5. Click **Review + Create**, wait for validation, then click **Create**.

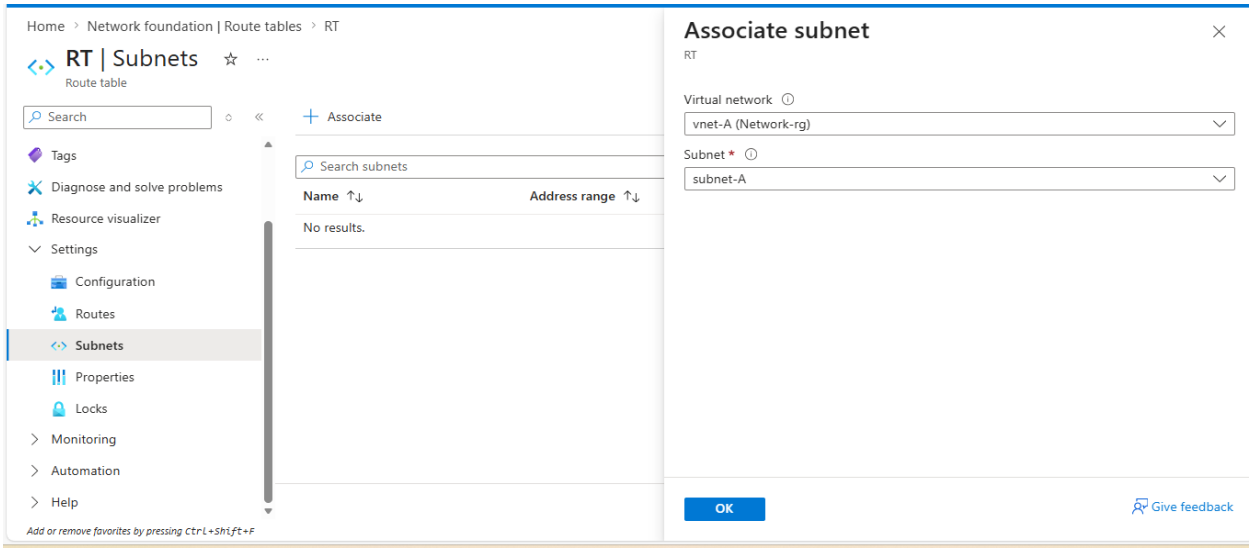
Step 2: Add Route

1. Open RT
2. Left Nevigation pane **settings > Routes → + Add**
3. Enter:
    - Route name: To-VNet-B
    - Destination Type: IP adresses
    - Address prefix: 10.1.0.0/16
    - Next hop type: **Virtual network**
4. Click **ADD**



Step 3: Associate Route Table to Subnet

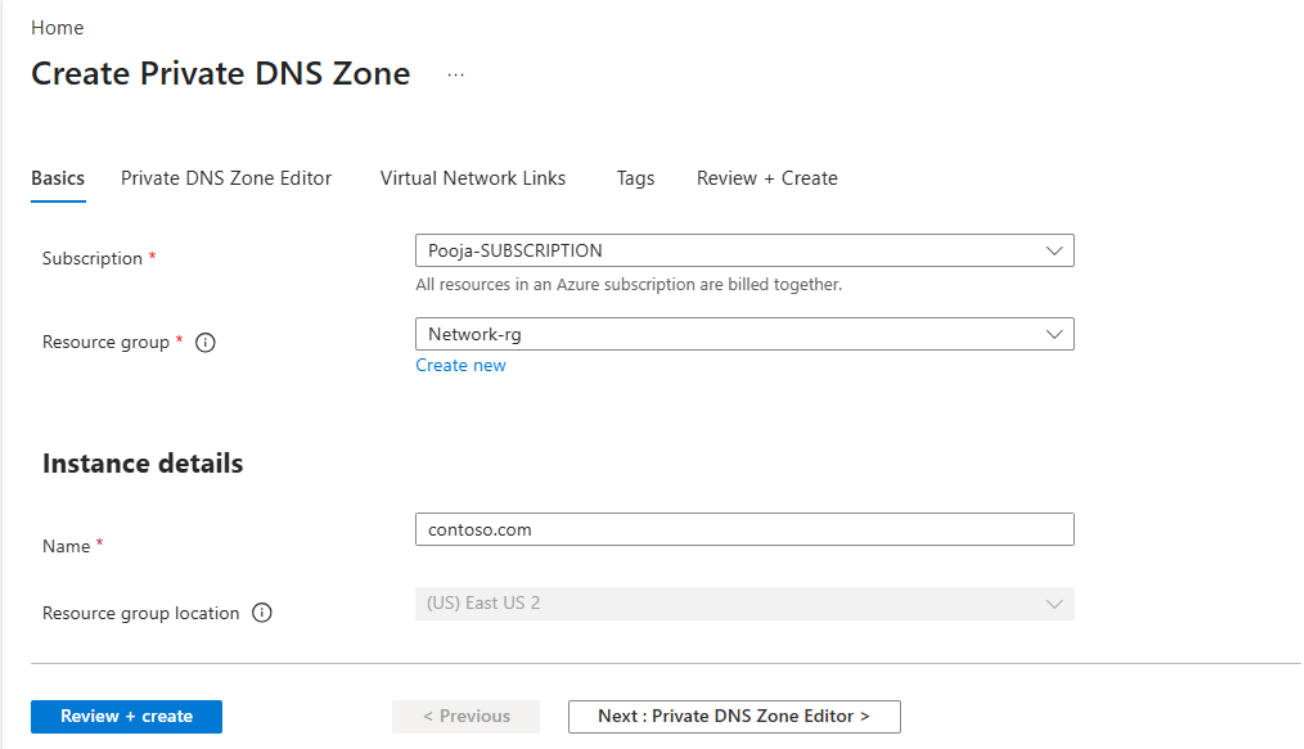
1. Go to your Route table RT
2. Left pane Settings > **Subnets**
3. Click **+ Associate**
4. Choose:
    - VNet-A
    - Subnet-A
5. Click **OK**



Exercise 4 – dns creation & record set

Step 1: Create Private DNS Zone

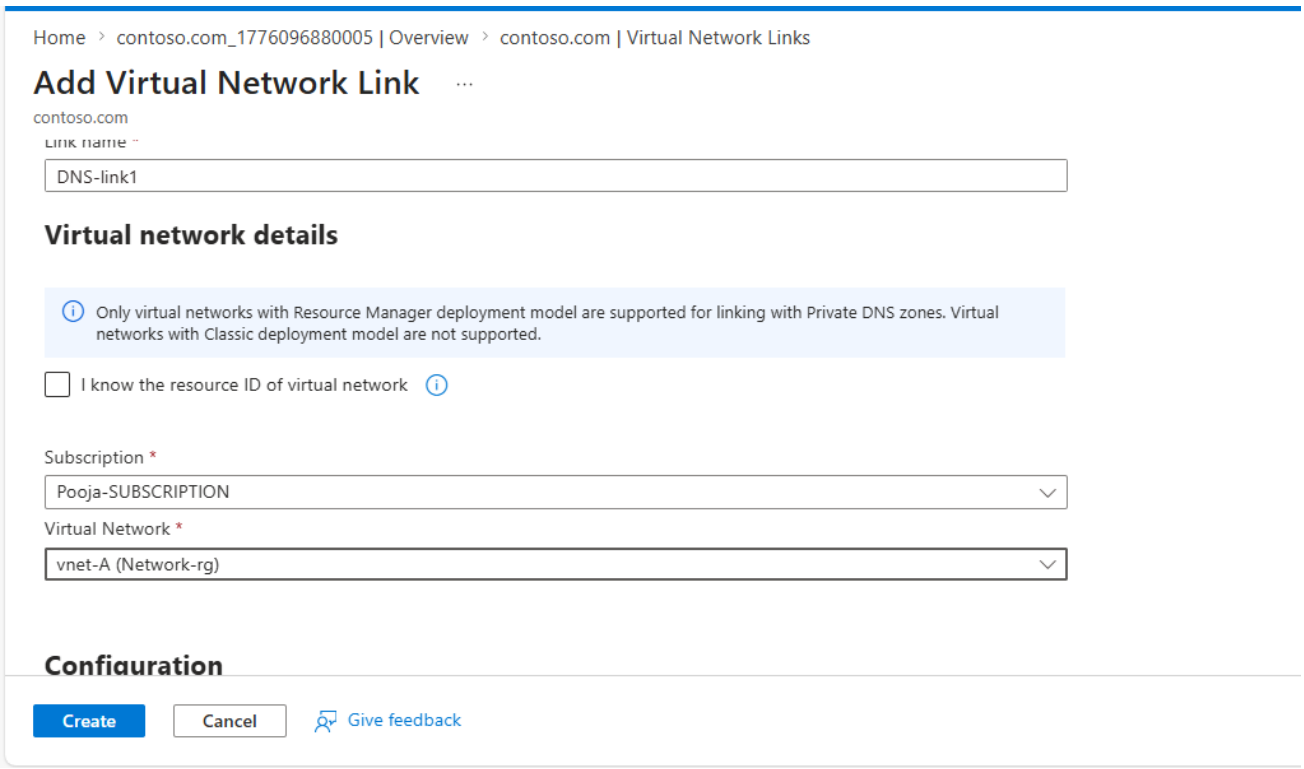
1. Go to **Private DNS Zones**
2. Click **+ Create**
3. Name: let say contoso.com
4. Resource Group: Choose yours



5. Click **Review + Create**, wait for validation, then click **Create**.

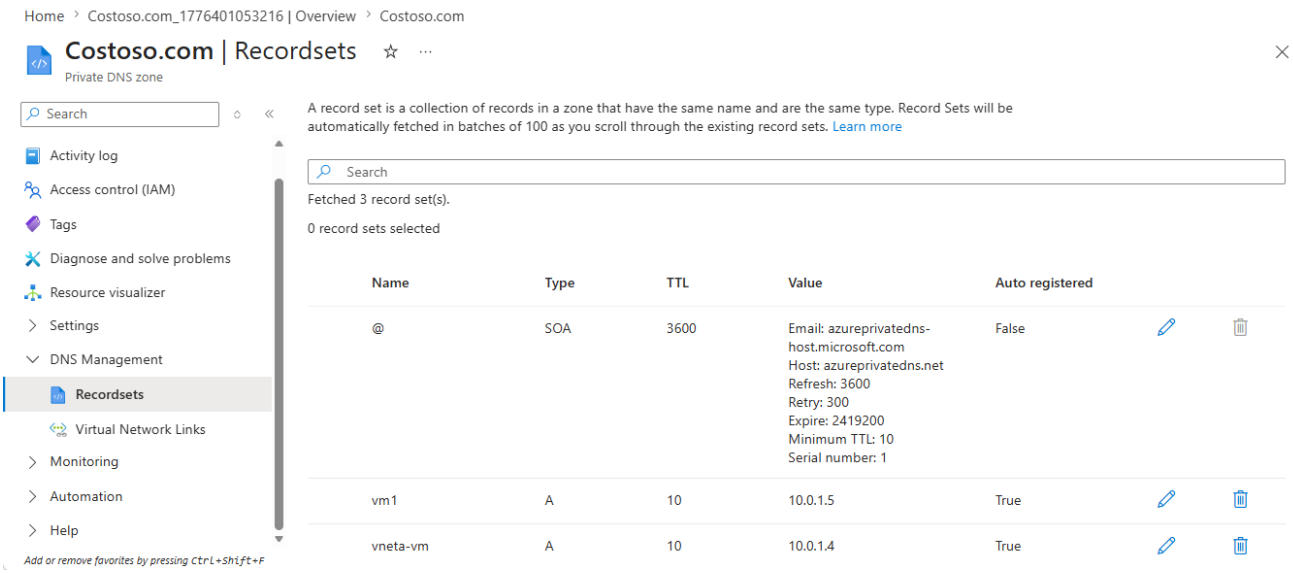
Step 2: Link DNS Zone to VNet

1. Open your DNS zone
2. Dns management > **Virtual network links**
3. Add:
    - Link name: DNS-link1
    - VNet: VNet-A
    - Scroll down **Configuration** -Auto registration: **Enabled** (means virtual machines created in the VNet are automatically registered with the Private DNS zone.)



Step 3: Add Record Set

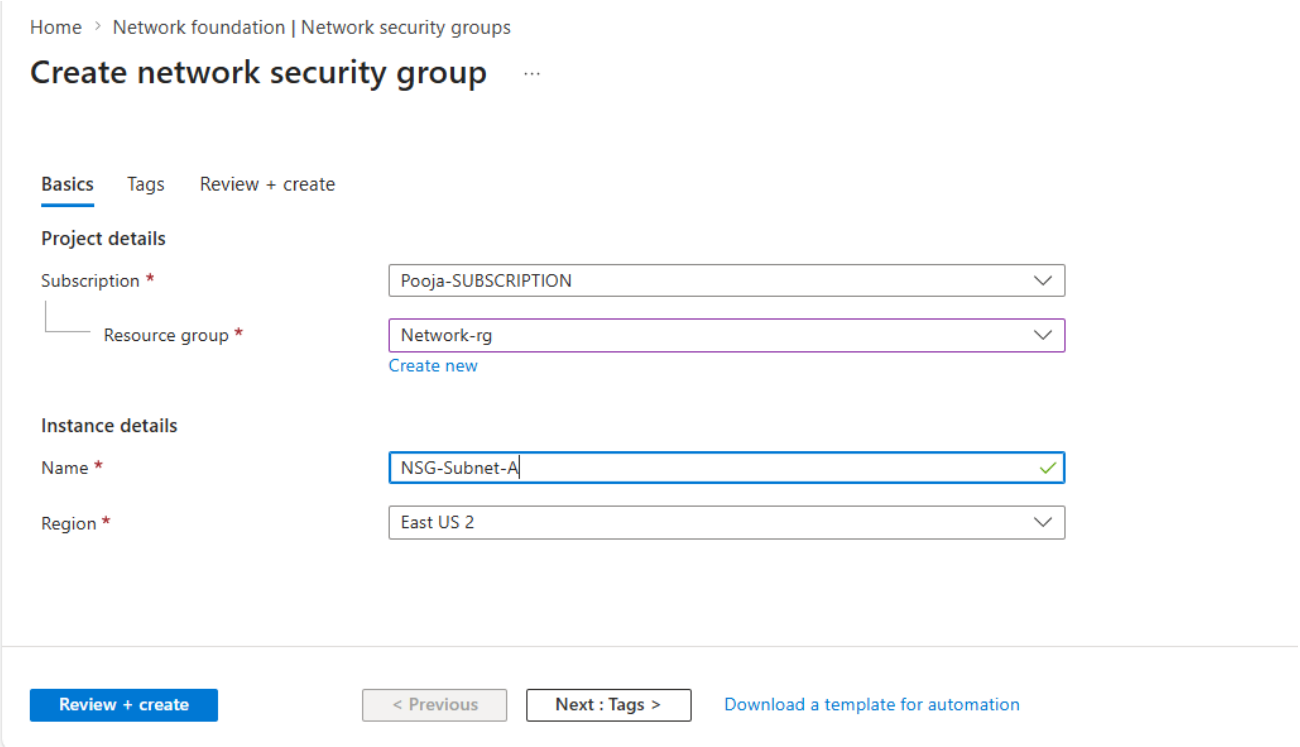
1. Click **+ Record Set** in the left pane
2. You can see record set have been created for VMs in vnet-A automatically
3. If you want to edit you can edit by clicking edit button, if you don't want any record set you can delete.



Exercise 5 – NSG creation and Associating NSG to subnet

Step 1: Create NSG

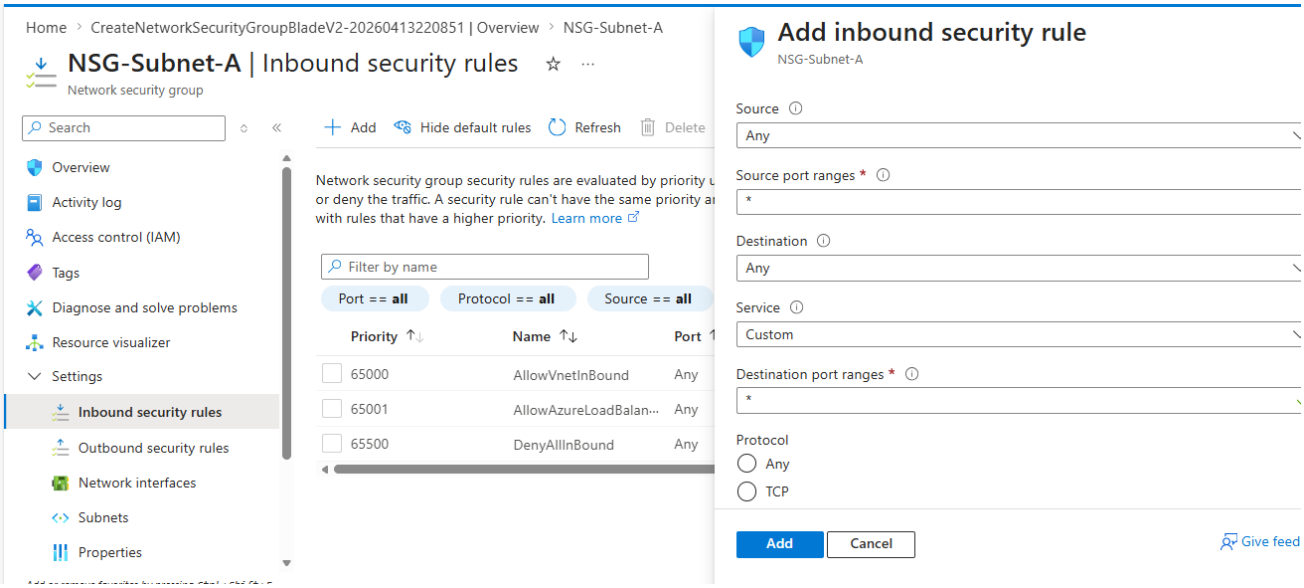
1. Go to **Network Security Groups**
2. Click **+ Create**
3. Name: NSG-Subnet-A
4. Resource Group: choose yours

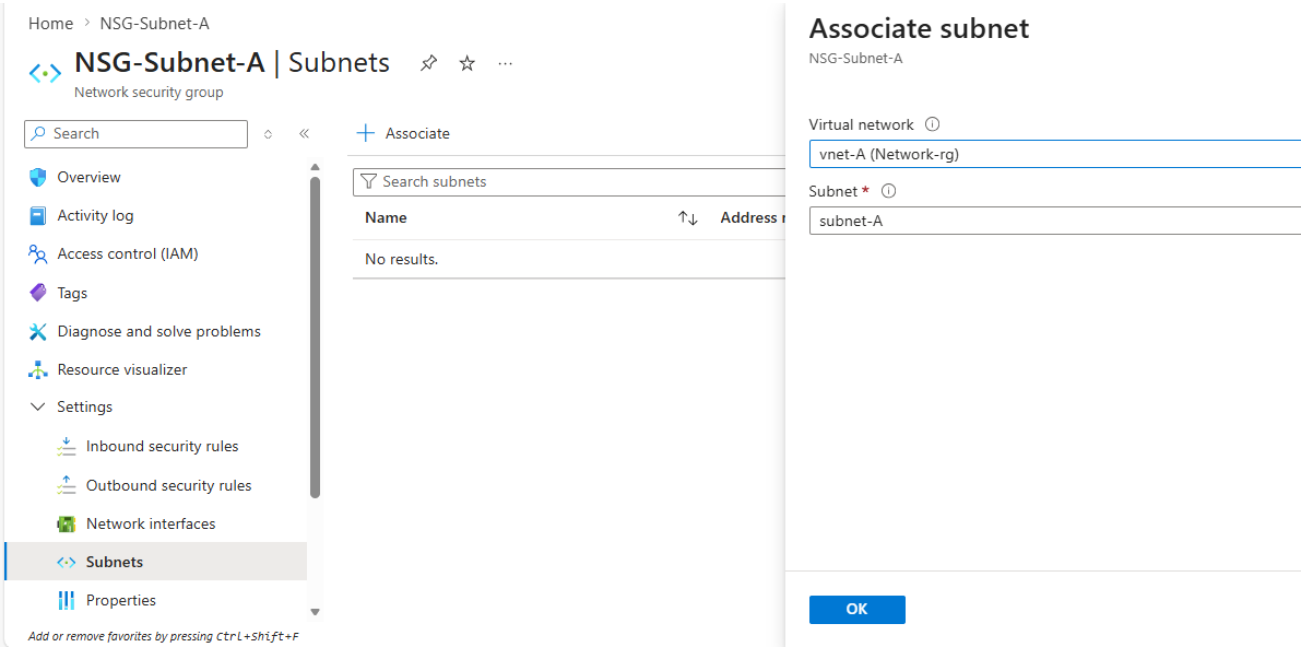


Click **Review + Create**, wait for validation, then click **Create**.

Step 2: Add Security Rule

1. Open NSG –
2. **Settings** > **Inbound security rules**
3. Click **+ Add**
4. Enter:
    - Source: Any
    - Protocol: ICMP
    - Action: Allow
    - Priority: 100
    - Name: Allow-Ping
5. Click **Add**



Step 3: Attach NSG to Subnet

1. NSG → **Settings** > **Subnets**
2. Click + Associate
3. Select:
    - VNet-A
    - Subnet-A
4. Click **OK**